Mock Midterm 2 For CSC182

Object-Oriented Programming in C++ — Practice Exam

Name: ..... ID: .....

Section: ..... Time: 90 minutes

Question 1: The code contains 10 errors, find and correct them:

```cpp
#include <iostream>
#include <string>
using namespace std;

class Student {
private:
string name;
double gpa;
Public:
Student(string n = "", double g) {
name = n;
gpa = g;
}
void setGpa(double g) {
gpa == g;
}
double getGpa() {
gpa;
}
String getName() {
return name;
}
void print() {
cout >> name >> " " >> gpa >> endl;
}
~Student(int x) {}
};

int main() {
Student s("Ali", 3.5);
s.gpa = 4.0;
cout << s.getName << endl;
print();
return 0;
}
```

Question 2: Class Employee

1- Create a class named Employee with private data members:

- **name of type string**
- **id of type int**
- **salary of type double**

2- Implement a parameterized constructor for the class that includes default values:

3- Implement the following public member functions:

- **Setter methods — assign employee data**
- **Getter methods — return employee data**
- **Method that returns the yearly salary (salary × 12)**
- **Print method — prints all employee information**

Question 3: Class Department

1- Create a class named Department with private data members:

- **name of type string**
- **emp1 of type Employee**
- **emp2 of type Employee**

2- Implement a parameterized constructor for Department:

3- Implement the following public member functions:

- Print method — prints all Department information
- Method to return the employee with the higher salary:
- Method that returns the average salary of both employees:
- Method that gives both employees a raise by a given percentage (e.g. *giveRaise(10)* increases each salary by 10%):

Question 4: main() Function

1. **Create two Employee objects:**
   - the first one using the parameterized constructor
   - the second one using the default values, then fill its data using the setter methods
2. **Create a Department object.**
3. **Print the information of the Department.**
4. **Show the employee with the higher salary.**
5. **Print the average salary.**
6. **Print the yearly salary of the first employee.**
7. **Give both employees a 10% raise, then print the Department information again.**

— Good Luck —

Answers

Question 1: The 10 errors and their corrections

#	Line	Wrong	Why	Correct
1	9	`Public:`	Keywords are case-sensitive	`public:`
2	10	`Student(string n = "", double g)`	Parameters after a default one must also have defaults	`Student(string n = "", double g = 0.0)`
3	15	`gpa == g;`	`==` compares, it does not assign	`gpa = g;`
4	18	`gpa;`	A `double` function must return a value	`return gpa;`
5	20	`String getName()`	The type is `string` (lowercase)	`string getName()`
6	24	`cout >> name >> ...`	`cout` uses the insertion operator `<<`	`cout << name << " " << gpa << endl;`
7	26	`~Student(int x) {}`	A destructor cannot take parameters	`~Student() {}`
8	31	`s.gpa = 4.0;`	`gpa` is private → use the setter	`s.setGpa(4.0);`
9	32	`s.getName`	A function call needs parentheses	`s.getName()`
10	33	`print();`	`print` is a member function → call it on the object	`s.print();`

Corrected code

```cpp
#include <iostream>
#include <string>
using namespace std;

class Student {
private:
    string name;
    double gpa;
public:
    Student(string n = "", double g = 0.0) {
        name = n;
        gpa = g;
    }
    void setGpa(double g) {
        gpa = g;
    }
    double getGpa() {
        return gpa;
    }
    string getName() {
        return name;
    }
    void print() {
        cout << name << " " << gpa << endl;
    }
    ~Student() {}
};

int main() {
    Student s("Ali", 3.5);
    s.setGpa(4.0);
    cout << s.getName() << endl;
    s.print();
    return 0;
}
```

Output: `Ali` then `Ali 4`

Question 2: Class Employee

1- Class Employee with private data members:

```cpp
#include <iostream>
#include <string>
using namespace std;

class Employee {
private:
    string name;
    int id;
    double salary;
```

2- Parameterized constructor with default values:

```cpp
public:
    // Parameterized constructor with default values
    Employee(string n = "No Name", int i = 0, double s = 0.0) {
        name = n;
        id = i;
        salary = s;
    }
```

> Since all parameters have default values, `Employee e;` is valid and creates an employee with name "No Name", id 0 and salary 0.

3- Setters, getters, yearly salary and print:

```cpp
    // Setters
    void setName(string n)   { name = n; }
    void setId(int i)        { id = i; }
    void setSalary(double s) { salary = s; }

    // Getters
    string getName()   { return name; }
    int    getId()     { return id; }
    double getSalary() { return salary; }

    // Return the yearly salary (12 months)
    double yearlySalary() {
        return salary * 12;
    }

    // Print all employee information
    void print() {
        cout << "Name  : " << name << endl;
        cout << "ID    : " << id << endl;
        cout << "Salary: " << salary << endl;
    }
};
```

Question 3: Class Department

1- Class Department with private data members:

```
class Department {
private:
    string name;
    Employee emp1;
    Employee emp2;
```

2- Parameterized constructor:

```
public:
    // Constructor using a member initializer list
    Department(string n, Employee e1, Employee e2) : name(n), emp1(e1), emp2(e2) {}
```

3- Member functions:

```
    // Print all department information
    void print() {
        cout << "Department: " << name << endl;
        cout << "--- Employee 1 ---" << endl;
        emp1.print();
        cout << "--- Employee 2 ---" << endl;
        emp2.print();
    }

    // Return the employee with the higher salary
    Employee highestPaid() {
        if (emp1.getSalary() > emp2.getSalary())
            return emp1;
        else
            return emp2;
    }

    // Return the average salary of both employees
    double averageSalary() {
        return (emp1.getSalary() + emp2.getSalary()) / 2;
    }

    // Give both employees a raise by a given percentage
    void giveRaise(double percent) {
        emp1.setSalary(emp1.getSalary() + emp1.getSalary() * percent / 100);
        emp2.setSalary(emp2.getSalary() + emp2.getSalary() * percent / 100);
    }
};
```

> Department cannot write `emp1.salary` because `salary` is private in Employee — it reads with `getSalary()` and changes it with `setSalary()`. Note: `(a + b) / 2` needs the parentheses, otherwise only `b` is divided by 2.

Question 4: main() Function

```cpp
int main() {
    // 1. Create two Employee objects
    Employee e1("Sara", 101, 9000);   // parameterized constructor
    Employee e2;                      // default values
    e2.setName("Omar");               // then fill it with setters
    e2.setId(102);
    e2.setSalary(12000);

    // 2. Create a Department object
    Department d("IT", e1, e2);

    // 3. Print the information of the department
    d.print();

    // 4. Show the employee with the higher salary
    cout << "\nHighest paid employee:" << endl;
    d.highestPaid().print();

    // 5. Print the average salary
    cout << "\nAverage salary: " << d.averageSalary() << endl;

    // 6. Print the yearly salary of the first employee
    cout << "Yearly salary of " << e1.getName() << ": " << e1.yearlySalary() << endl;

    // 7. Give both employees a 10% raise, then print again
    d.giveRaise(10);
    cout << "\nAfter a 10% raise:" << endl;
    d.print();

    return 0;
}
```

> Important: the Department stores **copies** of `e1` and `e2`. `d.giveRaise(10)` changes the copies inside `d` only — `e1` in main still has salary 9000.

Question 4: Program output

```
Department: IT
--- Employee 1 ---
Name  : Sara
ID    : 101
Salary: 9000
--- Employee 2 ---
Name  : Omar
ID    : 102
Salary: 12000

Highest paid employee:
Name  : Omar
ID    : 102
Salary: 12000

Average salary: 10500
Yearly salary of Sara: 108000

After a 10% raise:
Department: IT
--- Employee 1 ---
Name  : Sara
ID    : 101
Salary: 9900
--- Employee 2 ---
Name  : Omar
ID    : 102
Salary: 13200
```

Full Program (Questions 2 + 3 + 4 together)

Compiled and tested with g++ — copy it into any compiler (e.g. onlinegdb.com) to run it.

```cpp
#include <iostream>
#include <string>
using namespace std;

// ================= Class Employee =================
class Employee {
private:
    string name;
    int id;
    double salary;

public:
    // Parameterized constructor with default values
    Employee(string n = "No Name", int i = 0, double s = 0.0) {
        name = n;
        id = i;
        salary = s;
    }

    // Setters
    void setName(string n)   { name = n; }
    void setId(int i)        { id = i; }
    void setSalary(double s) { salary = s; }

    // Getters
    string getName()   { return name; }
    int    getId()     { return id; }
    double getSalary() { return salary; }

    // Return the yearly salary (12 months)
    double yearlySalary() {
        return salary * 12;
    }

    // Print all employee information
    void print() {
        cout << "Name  : " << name << endl;
        cout << "ID    : " << id << endl;
        cout << "Salary: " << salary << endl;
    }
};

// ================= Class Department =================
class Department {
private:
    string name;
    Employee emp1;
    Employee emp2;

public:
    // Parameterized constructor (member initializer list)
    Department(string n, Employee e1, Employee e2) : name(n), emp1(e1), emp2(e2) {}
```

Full Program (continued)

```
        // Print all department information
        void print() {
            cout << "Department: " << name << endl;
            cout << "--- Employee 1 ---" << endl;
            emp1.print();
            cout << "--- Employee 2 ---" << endl;
            emp2.print();
        }

        // Return the employee with the higher salary
        Employee highestPaid() {
            if (emp1.getSalary() > emp2.getSalary())
                return emp1;
            else
                return emp2;
        }

        // Return the average salary of both employees
        double averageSalary() {
            return (emp1.getSalary() + emp2.getSalary()) / 2;
        }

        // Give both employees a raise by a given percentage
        void giveRaise(double percent) {
            emp1.setSalary(emp1.getSalary() + emp1.getSalary() * percent / 100);
            emp2.setSalary(emp2.getSalary() + emp2.getSalary() * percent / 100);
        }
    };

    // ================= main() =================
    int main() {
        // 1. Create two Employee objects
        Employee e1("Sara", 101, 9000);   // parameterized constructor
        Employee e2;                      // default values
        e2.setName("Omar");               // then fill it with setters
        e2.setId(102);
        e2.setSalary(12000);

        // 2. Create a Department object
        Department d("IT", e1, e2);

        // 3. Print the information of the department
        d.print();

        // 4. Show the employee with the higher salary
        cout << "\nHighest paid employee:" << endl;
        d.highestPaid().print();

        // 5. Print the average salary
        cout << "\nAverage salary: " << d.averageSalary() << endl;

        // 6. Print the yearly salary of the first employee
        cout << "Yearly salary of " << e1.getName() << ": " << e1.yearlySalary() << endl;

        // 7. Give both employees a 10% raise, then print again
        d.giveRaise(10);
        cout << "\nAfter a 10% raise:" << endl;
        d.print();

        return 0;
    }
```